

Comparative Performance of Thirty Two Crops Grown in Antarctic Soil, Antarctic Moss and Peat MOSS Under Glass House Conditions in Antarctica

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Abstract

Eighty one varieties of 29 different vegetables, two varieties of grass and one variety of maize were grown in three different media viz. Antarctic soil, Antarctic moss and Peat moss. Performance of leafy vegetables (Spinach, fenugreek, lettuce, parseley, garlic, onion, mint and coriander), cucurbits (cucumber, bittergourd and squash), beans, brinjal, chilli and tomato was quite promising. Cruciferous plants (radish, knol khol, turnip, lai, cabbage and cauliflower) did not respond well during austral summer.

Introduction

The cold desert of Antarctica supports only an impoverished community of cold tolerant land plants that are capable of surviving lengthy winter months of total or near total darkness during which period photosynthesis can not take place. However, vascular flora in Antarctica consist of only three species of flowering plants (Green and Green, 1963). Growth must occur during short summer period which lasts for a few days to a few months depending on diverse factors like latitude, seasonal snow packs, elevation, topography etc. Moisture is the most important single variable provided mainly by atmospheric water vapour and by local melt supplies from snow, drift snow and permafrost. Stream run-off is exceedingly rare. Extreme cold, high winds and aridity inhibit growth even during 'summer' in most areas. Antarctic plants total about 800 species (Benton,1980) of which 350 are lichens and 100 are bryophytes. Numerous species of molds, yeast and other fungus as well as fresh water algae and bacteria are also found. These forms are extremely widespread and are reported from as far as 87 degree south latitude.

During the fifth Indian expedition to Antarctica, vegetables (spinach, cucumber, methi, tomato and radish) and ornamental plants were cultivated al Dakshin Gangotri station and Schirmacher Oasis in soil media (Joshi and

Banerjee,1988). During ninth and tenth expeditions a number of vegetables were cultivated in peat moss media and tomato through hydroponic system. To study the effect of complete light and absence of darkness on the performance of plants, different vegetables and crops were grown in different media during the eleventh expedition, under variable glass house conditions in Maitri station.

Materials and Methods

The glass house in Maitri station has a total area of 27.07 m² of which 41.52% area was utilised for agricultural experiments. The environmental condition of glass house was variable and the temperature was maintained by installing heat convector and water heating radiator. During the study period, soil temperature (5 cm depth) was located at ten locations inside the glass house. Atmospheric temperature (maximum and minimum) and relative humidity were recorded daily at 0800 and 1400 hrs. Light intensity was recorded at five places inside the glass house and at one place (helipad) outside the glass house at 0800 and 1400 hrs daily.

Different varieties of 29 horticultural crops (water melon, musk melon, bottlegourd, bittergourd, cucumber, summer squash, tomato, capsicum, chilli, onion, garlic, coriander, beet root, spinach, fenugreek, radish, lettuce, parseley, knol khol, turnip, cabbage, cauliflower, okra, brinjal, pea, french bean, carrot, mint and lai), two varieties of grass and one variety of maize were grown in three different media viz.,Antarctic moss (collected from the bank of Priyadarshini Lake), Antarctic soil (collected locally) and peat moss. FRP tray, wooden tray and plastic pots of different sizes were used for sowing and conducting the experiment.

Observations

Data on soil temperature, atmospheric temperature (maximum and minimum), relative humidity and light intensity of glass house are presented in Tables I, II,III and IV. Data on comparative performance of different parameters i.e. days for germination, percentage of germination, days for flowering, fruiting and marketable/harvestable stage, number of fruits, pods per plant, plant height and yield per plot and per hectare recorded are presented in Tables V, VI, VII and VIII.

Three varieties of tomato, one variety of coriander, two varieties each of cabbage and cauliflower, one variety, each of okra and pea, four varieties of carrot did not germinate in Antarctic soil whereas tomato (Pusa ruby), fenugreek (Pusa early bunches) and carrot (Early nantes) did not germinate in

Table I: Weekly Average Temperature (°C) of Soil Thermometer (5 cm) Recorded at 10 Locations Inside the Glass House In Antarctica

Location	Time Hrs	1-7 Jan 92	8-15 Jan 92	16-23 Jan 92	24-31 Jan 92	1-7 Feb 92	8-12 Feb 92
1	2	3	4	5	6	7	8
I	800	25.0(20.0-28.0)	25.25(18.0-31.0)	30.07(19.0-30.0)	21.87(18.0-30.0)	24.78(22.0-38.0)	25.16(22.5-27.0)
	1400	21.75(20.0-24.0)	22.75(18.5-27.0)	23.57(20.0-27.0)	20.00(16.0-28.0)	23.00(14.5-27.0)	20.5(20.0-21.0)
II	800	28.25(26.0-31.0)	25.25(19.6-31.5)	22.57(11.0-26.0)	18.38(12.5-32.0)	21.85(10.5-30.0)	20.5(19.5-24.0)
	1400	23.25(18.5-26.5)	20.75(18.0-21.5)	20.42(10.5-28.0)	18.64(15.0-25.0)	22.25(22.0-27.5)	17.00(15.5-18.5)
III	800	28.75(20.0-37.0)	29.81(19.0-30.5)	23.0(12.0-29.0)	16.12(11.0-23.0)	20.0(15.0-24.0)	19.66(17.0-20.0)
	1400	23.75(20.0-28.5)	25.06(18.5-31.0)	17.81(19.0-26.0)	17.28(14.0-22.0)	22.5(19.0-28.5)	15.75(15.8-16.0)
IV	800	20.75(19.0-26.0)	21.37(15.0-27.5)	19.0(16.0-25 .5)	18.50(12.5-32.0)	18.71(16.0-22.0)	19.66(14.0-23.0)
	1400	21.50(20.5-23.0)	20.43(17.0-23.0)	20.64(18.5-22.5)	15.78(14.0-28.0)	19.83(16.0-27.5)	17.50(17.0-18.0)
V	800	23.87(22.0-24.5)	20.50(16.0-25.0)	16.75(13.0-22.5)	23.7 5(12.0 -30.0)	20.21(16.0-26.0)	18.00(12.0-22.0)
	1400	21.50(18.5-21.5)	18.25(14.0-22.5)	17.42(14.0-20.5)	17.85(13.0-25.0)	19.91(16.5-22.0)	16.00(15.0-17.0)
VI	800	23.75(22.0-29.0)	19.75(18.0-26.00)	17.50(12.0-22.0)	16.25(14.0-29.0)	18.71(14.0-27.0)	20.66(17.0-21.0)
	1400	21.12(19.5-23.5)	20.00(16.0-23.0)	17.78(16.0-21.0)	15.07(12.0-24.0)	19.50(15.5-26.5)	17.00(16.0-18.0)
VII	800	25.87(23.0-29.0)	22.80(24.0-30.0)	18.00(15.0-28.0)	15.68(13.0-28.0)	19.35(12.0-24.0)	18.33(15.0-20.0)

Comparative Performance of Thirty Two Crops Grown...

Contd.

Table I — *Contd*

VIII	1400	22.00(21.0-24.0)	20.50(17.5-20.0)	17.21(15.0-22.5)	16.35(12.0-24.0)	19.75(18.0-22.0)	16.50(15.0-18.0)
	800	23.62(19.0-27.0)	23.25(15.0-30.00)	17.75(12.0-21.0)	15.98(9.5-35.0)	16.07(14.0-24.0)	22.00(19.0-24.0)
IX	1400	20.50(16.0-24.0)	20.56(17.0-23.5)	18.57(16.0-20.0)	15.14(12.0-21.0)	16.83(15.0-18.0)	17.5(16.5-18.5)
	803	24.50(19.0-28.5)	26.65(19.5-31.0)	18.62(12.0-27.5)	18.18(10.0-38.0)	20.71(16.0-27.0)	21.83(20.0-25.0)
X	1400	22.00(21.0-23.0)	20.50(16.0-23.0)	20.00(16.5-22.0)	17.42(13.0-25.0)	19.50(17.5-22.0)	18.75(18.5-19.0)
	800	26.00(23.0-31.0)	25.25(18.0-31.0)	21.37(12.0-29.0)	19.81(12.0-33.0)	22.28(16.0-28.0)	19.00(18.0.20.0)
	1400	21.12(18,0-24.0)	21.5(18.0-22.0)	21.23(18.0-26.0)	19.85(16.0-27.0)	23.66(20.0-24.0)	16.25(16.0-16.5)

Table II: Average Maximum and Minimum Temperature Inside the Glass House in Antarctica

Thermometer	Time Hrs	15-20 Jan 92	21-27 Jan 92	28 Jan - 3 Feb 92	4-10 Feb 92	11-13 Feb 92
Maximum	1400	38.42(25.0-50.0)	33.71(28.0-38.0)	30.00(25.0-37.0)	34.57(28.0-42.0)	34.66 (33.0-37.0)
Minimum	800	25.42(20.0-30.0)	24.28(19.0-31.0)	21.14(18.0-28.0)	21.57(16.0-27.0)	25.00(24.0-26.0)

Table III: Weekly Average Humidity (%) Inside the Glass House in Antarctica

15-20 Jan92	21-27 Jan 92	28 Jan-3 Feb 92	4-10 Feb 92	11-13 Feb 92
29.09	43.26	54.27	53.00	38.33
(10.0-42.5)	(12.0-64.0)	(25.0-88.0)	(18.5-82.5)	(15.0-50.0)

Table IV: Weekly Average Light Intensity (Lux) at 5 Places Inside the Glass House and at Helipad (Outside the Glass House) In Antarctica

Location	Time Hrs	1-7 Jan 92	8-15 Jan 92	16-23 Jan 92	24-31 Jan 92	1-7 Feb 92	8-12 Feb 92
1 (North)	800	18690 (2670-30000)	9792 (5800-33300)	3437.5 (1235-4900)	4001.42 (4220-11560)	4461.57 (1341-6240)	375.3.33 (2740-5040)
	1400	2200 (600-4200)	4627.14 (1200-10000)	1782.16 (1150-2910)	2011.00 (1500-2660)	1869.33 (1296-3790)	190 5 (1430-2380)
2 (South)	800	9825 (1100-18000)	17757 (9800-82000)	6705 (3070-14400)	6492.85 (3970-14280)	12628.57 (4650-21800)	5896.66 (5290-6630)
	1400	1393.33 (1180-1700)	3465 (1800-21500)	3302.33 (2060-4630)	3125 (1165-5190)	3400 (1920-4420)	3400 (2530-4570)
3 (East)	800	5575 (4600-6500)	14921.25 (5270-53400)	4640 (2020-7740)	4428.57 (1540-12080)	4628.57 (2990-38300)	4003.33 (3290-4740)
	1400	1466.66 (1000-2000)	7607.14 (1600-20600)	2783.5 (1270-5450)	2701.66 (1362.3610)	2556.66 (1880-3300)	3205 (1950-4460)
4 (West)	800	18712.5 (9200-53200)	16011.00 (5800-35700)	19415(3270-36700)	18900.00(5020-74300)	25730.0 (8400-86800)	12850 (4280-26800)
	1400	1833.33 (1000-2300)	6315 (3300-17100)	4121.66 (2530-8570)	3886 (1910-5660)	4532.50 (3170-6340)	3920 (2670-4170)

Comparative Performance of Thirty Two Crops Grown...

(Contd.)

Table IV — *Contd.*

5 (Middle portion)	800	6700 (4200-13600)	18171.25 (8600-52300)	9865 (3550-26700)	16777.50 (3240-93800)	21011.14 (6240-37100)	12250 (6640-21900)
	1400	2383.33 (2050-3000)	7085.71 (1870-26800)	3556.66 (2760-6630)	4347.16 (1823-6480)	4455 (2690-6440)	3677 (2700-4640)
Helipad	800	107900 (90000-125000)	128400 (56500-140400)	45142.5 (1860-111800)	49483.33 (20300-123500)	144840 (84100-135800)	34400 (25300-43500)
	1400	124400 (120000 128800)	120750 (67500-145200)	97200 (90100-97200)	60566.66 (28300-98200)	120620 (98200-132500)	24050 (18300-29800)

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Table V: Performance of Germination in Different Media

SI. No.	Vegetable	Variety	Peat moss		Antarctic soil		Antarctic moss	
			Days to 50% Germination	Germination percentage	Days to 50% Germination	Germination percentage	Days to 50% Germination	Germination percentage
1	2	3	4	5	6	7	8	9
1.	Water melon	Ay	7	90	7	80	5	90
2	Musk melon	-	5	100	5	90	4	100
3.	Bottlegourd	PSPL	10	80	8	90	7	60
4.	Bottlegourd	Pusa Navin	7	80	8	60	6	50
5.	Bottlegourd	Arka Harit	7	70	10	70	10	40
6.	Cucumber	Japanese Long green	5	100	6	70	4	100
7.	Cucumber	Pasa Sanyog	5	100	6	90	4	100
8.	Summer squash	Pusa Alankar	6	100	8	80	8	70
9.	Summer squash	Australian green	6	100	6	100	7	90
10 .	Bittergourd	2ft long Jaunpuri	12	60	14	30	12	40
11.	Bittergourd	White long Coimbatoret	10	80	13	40	12	50
12.	Bittergourd	Pusa Viresh	10	70	11	30	10	60

Comparative Performance of Thirty Two Crops Grown...

(Contd.)

Table V — *Contd.*

13.	Bittergourd	Pusa Domausami	10	70	11	40	8	40
14.	Tomato	Arka Saurabh	7	40	Not germinated		6	40
15.	Tomato	Best of All India	7	50	8	20	8	30
16.	Tomato	VL-2	6	100	12	20	9	70
17.	Tomato	Marglobe	8	40	11	20	9	70
18.	Tomato	Pusa Ruby	Not germinated		11	50	Not germinated	
19.	Tomato	Pusa Gaurav	8	80	11	20	8	80
20.	Tomato	HT-8	8	40	Not germinated		Not germinated	
21.	Tomato	HT-6	8	40	Not germinated		Not germinated	
22.	Tomato	Pusa Sheetal	9	70	13	20	10	10
23.	Capsicum	California wonder	9	100	13	20	10	50
24.	Capsicum	Yellow wonder	9	50	13	30	11	30
25.	Capsicum	HC-202	10	40	14	40	12	50
26.	Chilli	Green long Jwala	11	60	13	30	12	40
27.	Chilli	Pant C-1	10	40	13	20	12	30

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Table V — *Contd.*

28.	Chilli	Pusa Jwala	10	50	11	20	13	30
29.	Onion	Nasik Red	10	40	11	20	8	10
30.	Onion	Pusa Red	10	30	11	20	8	10
31.	Coriander	Bulgarian	7	90	Not germinated		11	10
32.	Beet root	Red Ball	8	80	10	90	6	30
33.	Beet root	Detroit Dark red	5	100	5	100	5	100
34.	Beet root	Crimpson globe	5	100	5	80	5	100
35.	Spinach	Palak Banarasi	6	100	9	100	7	100
36.	Spinach	Pusa Hari	5	100	5	90	4	90
37.	Methi	Pusa Early Bunches	Not terminated		13	10	9	60
38.	Methi	Kashmiri	4	50	3	70	4	90
39.	Methi	Pusa Kasuri	3	80	2	25	2	100
40.	Lai	ARU Black	6	90	8	20	5	20
41.	Radish	White ICEL	4	90	3	80	9	90
42.	Radish	Punjab Safed	4	100	3	80	4	100

Comparative Performance of Thirty Two Crops Grown...

(Contd.)

Table V — *Contd.*

43.	Radish	Pusa Himani	4	100	3	60	3	90
44.	Radish	Scarlet long	4	100	5	90	4	100
45.	Radish	Silk Pak	4	100	5	80	3	100
46.	Radish	Leh local	3	100	5	90	3	100
47.	Lettuce	CO5	7	30	11	10	6	20
48.	Lettuce	Chinese yellow	7	80	5	60	4	70
49.	Lettuce	Great lake	4	60	5	30	3	60
50.	Lettuce	Cabbage	10	20	13	10	13	20
51.	Parseley	Moss curled	11	30	13	30	11	60
52.	Knol-khol	White Vienna	3	80	5	80	4	50
53.	Knol-khol	Early W.V	5	50	5	20	9	60
54.	Knol-khol	E.V.W.	3	20	5	20	8	40
55.	Knol-khol	King of Market	6	60	5	10	6	10
56.	Knol-khol	Goliath	-	-	-	-	-	-
57.	Turnip	P.T.W Globe	3	100	5	30	4	70

(Contd).

Table V — *Contd.*

58.	Turnip	Pusa Sawni	8	60	5	40	8	80
59.	Turnip	Golden Ball	10	25	6	10	8	50
60.	Cabbage	Pusa Mukta	4	90	Not germinated		3	80
61.	Cabbage	Golden Acre	4	60	Not germinated		3	50
62.	Cabbage	Drum Head	4	80	Not germinated		4	80
63.	Cauliflower	PSB-1	4	90	Not germinated		5	40
64.	Cmliftower	Pusa Snow Ball K-I	4	80	9	10	5	80
65.	Cauliflower	Super Snow Ball	4	90	9	30	4	100
66.	Okra	Gumbo	5	80	Not germinated		6	40
67.	Brinjal	PPL	8	10	13	40	8	50
68.	Pea	Bonneville	6	90	Not germinated		5	100
69.	Pea	Arkel	5	too	11	80	4	100
70.	Pea	Azad-I	6	100	11	60	6	100
71.	Bean	High Alt. Sel	5	100	11	60	5	50
72.	Bean	Master Piece	6	100	8	80	6	70

Comparative Performance of Thirty Two Crops Grown...

(Contd.)

Table V—*Contd.*

73.	Bean	VL-Bauni	6	100	10	60	8	100
74.	Bean	Pant Anupma	6	100	10	30	6	40
75.	Bean	Pusa Cheese contender	6	100	6	100	10	100
76.	Bean	Pusa Parvati	6	100	6	100	8	100
77.	Bean	Asparagus Black seeded	5	100	6	100	5	100
78.	Carrot	Pusa Yomed Gainy	6	80	Not germinated		8	90
79.	Carrot	Pusa Kesar	6	100	Not germinated		5	40
80.	Carrot	Pusa Meghali	7	20	Not germinated		5	40
81.	Carrot	Early Nantes	Not germinated		Not germinated		8	40
82.	Lucern	Local	6	50	9	20	4	50
83.	Cocks foot	Dactylis	6	90	9	40	6	80
84.	Maize	Auli composit	4	100	4	100	4	100

Table VI: Performance on Flowering, Fruiting and Maturity in Different Media											
Sl. No.	Vegetables	Variety	Peat moss			Antarctic soil			Antarctic moss		
			Days to 50% flowering	Days to fruiting	Days to harvesting	Days to 50% flowering	Days to fruiting	Days to harvesting	Days to 50% flowering	Days to fruiting	Days to harvesting
1	2	3	4	5	6	7	8	9	10	11	12
1.	Water melon	Ay	-	-	-	-	-	-	-	-	-
2.	Musk Melon	-	34	-	-	-	-	-	-	-	-
3.	Bottlegourd	PSPL	-	-	-	-	-	-	-	-	-
4.	Bottlegourd	Puia Navin	51	-	-	-	-	-	-	-	-
5.	Bottlegourd	Arka Harit	-	-	-	-	-	-	-	-	-
6.	Cucumber	Japanese long Green	33	35	-	-	-	-	-	-	-
7.	Cucumber	Pusa Sanyog	31	33	-	-	-	-	-	-	-
8..	Summer Squash	Pusa Alankar	25	40	-	-	-	-	-	-	-
9.	Summer Squash	Australian green	26	40	-	-	-	-	-	-	-

Comparative Performance of Thirty Two Crops Grown...

(Contd.)

Table VI — *Contd.*

10.	Bittergourd	2 ft. long Jaunpuri	49	-	-	-	-	-	-	-	-
11.	Bittergourd	White long Coimbatore	45	48	-	-	-	-	-	-	-
12.	Bittergourd	Pusa Viresh	-	-	-	-	-	-	-	-	-
13.	Bittergourd	Pusa Domausami	45	48	-	-	-	-	-	-	-
14.	Tomato	Arka Saurabh	-	-	-	-	-	-	-	-	-
15.	Tomato	Best of All India	-	-	-	-	-	-	-	-	-
16.	Tomato	VL-2	48	-	-	-	-	-	-	-	-
17.	Tomato	Marglobe	48	-	-	-	-	-	-	-	-
18.	Tomato	Pusa Ruby	-	-	-	-	-	-	-	-	-
19.	Tomato	Pusa Gaurav	-	-	-	-	-	-	-	-	-
20.	Tomato	HT-8	51	-	-	-	-	-	-	-	-
21.	Tomato	HT-6	-	-	-	-	-	-	-	-	-
22.	Tomato	Pusa Sheetal	-	-	-	-	-	-	-	-	-
23.	Capsicum	California wonder	-	-	-	-	-	-	-	-	-

(*Contd.*)

Table VI -- *Contd.*

24.	Capsicum	Yellow wonder	-	-	-	-	-	-	-	-	-
25.	Capsicum	HC-202	-	-	-	-	-	-	-	-	-
26.	Chilli	Green long Jwala	-	-	-	-	-	-	-	-	-
27.	Chilli	Pant C-1	-	-	-	-	-	-	-	-	-
28.	Chilli	Pusa Jwala.	-	-	-	-	-	-	-	-	-
29.	Onion	Nasik red	-	-	-	-	-	-	-	-	-
30.	Onion	Pusa red	-	-	-	-	-	-	-	-	-
31.	Coriander	Bulgarian	-	-	26	-	-	-	-	-	-
32.	Beetroot	Red ball	-	-	26	-	-	-	-	-	-
33.	Beet root	Detroit Dark Red	-	-	26	-	-	-	-	-	-
34.	Beetroot	Crimpsn Globe	-	-	26	-	-	-	-	-	-
35.	Spinach	Palak Banarasi	-	-	30	-	-	-	-	-	-
36.	Spinach	Pusa Hari	-	-	38	-	-	-	-	-	-
37.	Methi	Pusa Early Bunches	-	-	-	-	.	-	-	-	-
38.	Methi	Kashmiri	38	-	32	-	-	-	-	-	-

(Contd.)

Comparative Pmfonumnee of Thirty Two Cmps Grown...

Table VI — *Contd*

39.	Methi	Pusa Kasuri	40	-	32	-	-	-	-	-	-
40.	Lai	ARU Black	After 12 days growth was poor and plants got dried								
41.	Radish	White ICEL	Upto 40 days there was no root formation and plant bears four leaves								
42.	Radish	Punjab Safed	-	-	-	-	-	-	-	-	-
43.	Radish	Pusa Himani	-	-	-	-	-	-	-	-	-
44.	Radish	Scarlet long	-	-	-	-	-	-	-	-	-
45.	Radish	Silk pak	-	-	-	-	-	-	-	-	-
46.	Radish	Leh local	-	-	-	-	-	-	-	-	-
47.	Lettuce	COS	-	-	30	-	-	-	-	-	-
48.	Lettuce	Chinese Yellow	-	-	28	-	-	-	-	-	-
49.	Lettuce	Great lake	-	-	24	-	-	-	-	-	-
50.	Lettuce	Cabbage	-	-	36	-	-	-	-	-	-
51.	Parseley	Moss Curled	-	-	40	-	-	-	-	-	-
52.	Knol-khol	White Vienna	}	After transplantation, growth was very poor and 18 days after transplanting maximum plants died.							
53.	Knol-khol	Early W.V.									
54.	Knol-khol	E.V.W.									

Contd.

Table VI — *Contd.*

55.	Knol-khol	King of Market	}	After transplantation, growth was very poor and 18 days after transplanting maximum plants died							
56.	Knol-khol	Goliath									
57.	Turnip	P.T.W. Globe									
58.	Turnip	Pusa Sawni									
59.	Turnip	Golden Ball									
60.	Cabbage	Pusa Mukta	}	After 12 days of transplanting, the plants started bolting with poor growth.							
61.	Cabbage	Golden Acre									
62.	Cabbage	Drum Head									
63.	Cauliflower	PSB-1									
64.	Cauliflower	Pusa Snow Ball K-I									
65.	Cauliflower	Super Snow Ball									
66.	Okra	Gumbo		Very slow growth	-	-	-	-	-	-	-
67.	Brinjal	PPL		-	-	-	-	-	-	-	-
68.	Pea	Bonneville		37	39	-	-	-	37	39	-
69.	Pea	Arkel		38	40	dried	-	-	38	40	-

Comparative Performance of Thirty Two Crops Grown...

Contd.

Table VII: Days to Germination, Flowering, Fruiting, Fruit and Pod Number and Their Size, Plant Height of Different Vegetables and Grasses in Peat Moss Media (Plant height was recorded on 51st day i.e. 21st Feb 92)

Sl.No.	Variety	Sowing date	Days to 50% germination	Days to transplanting	Days to flowering	Days to fruiting	Fruit/pod (No.) per plant (Av)	Average size of fruit/pod per plant(cm)	Average height of plant (cm)
1.	Bean (High Altitude Selection)	31.12.91	5	-	43	51	18.0	4.5	157.48
2.	Bean (Asparagus)	31.12.91	6	-	-	-	-	-	12.70
3.	Bean (Pu&a Cheese Contender)	31.12.91	6	-	33	43	9.0	7.0	17.78
4.	Bean (Master Piece)	31.12.91	6	-	33	39	22.5	8.4	33.5
5.	Besti (Pusa Parvati)	31.12.91	6	-	35	44	10.0	8.0	32.0
6.	Bean (Pant Anupma)	31.12.91	6	-	35	39	12.0	9.0	34.0
7.	Bean (VL Bauni)	31.12.91	6	-	37	47	6.5	6.0	28.0
8.	Cucumber (Pusa Sanyog)	4.12.91	5	28	35	37	7.0	3.5*	104.14
9.	Cucumber (Pusa Sanyog)	31.12.91	5	14	31	33	8.0	3.0	43.18
10.	Cucumber (Japanese Long Green)	31.12.91	5	24	33	35	6.0	3.0	91.44
11.	Bottlegourd (Pusa Navin)	31.12.91	7	14	51	-	-	-	170.18
12.	Bottlegourd (PSPL)	31.12.91	10	14	-	-	-	-	138.50
13.	Bittergourd (Pusa Domausami)	31.12.91	10	14	45	48	-	-	144.78

Comparative Performance of Thirty Two Crops Grown...

Contd.

Table VII — *Contd.*

14.	Bittergourd (2 ft. long Jaunpuri)	31.12.91	10	14	49	-	-	-	81.28
15.	Muskmelon (Pusa Madhuras)	31.12.91	5	14	34	-	-	-	170.18
16.	Tomato (VL-2)	31.12.91	8	14	48	-	-	-	35.56
17.	Tomato (Marglobe)	31.12.91	8	14	48	-	-	-	22.86
18.	Tomato (HT-6)	31.12.91	8	14	51	-	-	-	20.32
19.	Tomato (HT-8)	31.12.91	8	14	-	-	-	-	15.32
20.	Capsicum (California Wonder)	31.12.91	9	18	-	-	-	-	17.78
21.	Capsicum (HC-202)	31.12.91	10	18	-	-	-	-	10.16
22.	Chilli (Pant C-1)	31.12.91	10	20	-	-	-	-	7.62
23.	Chilli (P. Jwala)	31.12.91	10	21	-	-	-	-	6.35
24.	Okra (Gumbo)	31.12.91	5	16	-	-	-	-	6.35
25.	Lucern	31.12.91	6	-	41	-	-	-	35.56
26.	Cocks foot grass (<i>Dactylis glomerata</i>)	31.12.91	6	-	-	-	-	-	17.78
27.	Maize (Auli composite)	31.12.91	4	-	-	-	-	-	35.56

* 30 cm length of cucumber on 65th day (communicated on ship from Maitri)

Table VIII: General Observations of Leafy Vegetables Grown in Peat Moss Media												
SI No.	Vegetables	Sowing date	Days to 50% germination	Av. plant height on 43rd day (cm)	Av. No. of leaves per plant	Av. weight per plant (g)	Av. length of leaf (cm)	Yield in 24x43 cm tray (g)	Yield (q/ha)	Days on Ist harvest	Days on IInd harvest	Days on Illrd harvest
1.	Methi (Pusa Kasuri)	31.12.91	02	28.00 (22.0-32.0)	41.20 (24.0-73.0)	12.00 (8.0-18.0)	-	390	377.70	32	40	-
2.	Lettuce (Chinese yellow)	31.12.91	07	-	5.50	24.50	-	240	232.55	30	34	40
3.	Lettuce (Great Lake)	31.12.91	04	-	6.50	26.50	-	320	310.07	28	32	38
4.	Spinach (Pusa Hari)	31.12.91	05	-	5.80	30.00	-	380	368.21	30	33.	37
5.	Spinach (Palak Banarasi)	31.12.91	06	-	5.00	24.00	-	290	281.00	30	34	43
6.	Coriander (Bulgarian)	31.12.91	07	-	-	28.00	-	-	27.13	26	-	-
7.	Beet Root (Detroit Dark Red)	31.12.91	05	-	5.00	22.50	-	300	290.69	26	38	43
8.	Beet Root (Red ball)	31.12.91	08	-	4.50	07.60	-	265	256.78	26	38	43
9.	Garlic (DARL-52)	30.12.91	04	-	5.00	12.50	30.50 (20.0-32.0)	150	145.34	26	34	-
10.	Onion Bulb (Nasik Red)	30.12.91	05	-	16.50	45.00	32.50 (26.0-43.0)	350	339.14	26	34	

Comparative Performance of Thirty Two Crops Grown...

peat moss media (Table V). In Antarctic moss only three varieties of tomato did not germinate. When compared to each other, a number of varieties germinated late in all three media. Water melon, musk melon, cucumber, one variety each of beet root and spinach, one variety of fenugreek, radish, lettuce, lucern, cocks foot, two varieties each of onion and bottlegourd germinated earlier in Antarctic moss media. Similarly, two varieties each of fenugreek, radish, knol khol and cabbage germinated earlier in Antarctic soil compared to Antarctic moss and peat moss media. The performance of cruciferous crops viz., radish, lai, knol khol, turnip, cabbage and cauliflower was very poor during the study period (austral summer) in Antarctica. After twelve days of germination lai got dried. Similarly, knol khol, turnip, cabbage and cauliflower started bolting 12 days after transplantation and root formation was not observed in radish till 40th day of sowing. Poor growth was observed in okra.

The earliest flowering was recorded in summer squash i.e. in Pusa Alankar variety on the 25th and in Australian green variety (Fig 14) on the 26th day and in cucumber (Pusa Sanyog) on the 31st day of sowing (Table VI). Flowering in musk melon, bottlegourd, cucumber (Japanese long green), tomato (VL-2, Marglobe and HT-8), bittergourd 2ft. long Jaunpuri, white Coimbatore and Pusa Domausami, fenugreek Kashmiri and Pusa Kasuri, pea Bonneville, Arkel and Azad and bean high altitude selection, master piece, VL Bauni-1, Pant Anupama, Pusa cheese, Contender and Pusa Parvati, was recorded on 34,51,33,48,48,51,49,45,45,38,40,37,38,32,43,32,37,37,35,33 and 35th days of sowing in peat moss media, respectively whereas in Antarctic moss media, the flowering was recorded on 37,38,32,33,37,35,33 and 35th day in pea Bonneville, Arkel and Azad variety, in Bean Master Piece, VL Bauni-1, Pant Anupama, Pusa cheese contender and Pusa Parvati variety, respectively (Table VI). The flowering days in Antarctic moss and peat moss media did not differ. First fruiting was observed in cucumber (Pusa Sanyog) on 33rd day of sowing followed by pea (Azad) and cucumber (Japanese long green) on 35th day in peat-moss media. Fruiting started on 40th day in summer squash (Fig 14), on 48th day in Pusa Domausami variety of bittergourd (Figs 10 and 11), on 30th, 40th and 35th day in Bonneville, Arkel and Azad variety of pea, respectively; on 51st day in bean (High altitude selection), on 39th day in Pusa Parvati and Master piece and on 40th day in VL Bauni, Pant Anupama and Pusa cheese contender variety of bean in peat moss. In comparison, in Antarctic moss media, fruiting was observed on 39th, 40th, 34th, 39th, 42nd, 41st, 42nd and 40th day of sowing in Bonneville, Arkel and Azad varieties of pea and Master piece (Fig 22), VL Bauni, Pant Anupama, Pusa cheese contender (Figs 16 and 17) and Pusa Parvati variety of bean, respectively. Asparagus black seeded variety of bean did neither flower nor fruit in any media till 51st day of sowing. The performance of bean in Antarctic moss was found to be as good as in peat moss

PLATE 1



Fig 1 : Spinach variety Pusa Hari in peat moss media



Fig 2 : Fenugreek variety Kasuri in peat moss media

PLATE 2



Fig 3: Coriander variety Bulgarian in peat moss media.



Fig 4: Beetroot variety Detroit dark Red and Red Ball

PLATE 3



Fig 5: Lettuce variety Great Lake and Chinese Yellow.



Fig 6 : Lettuce variety Cabbage

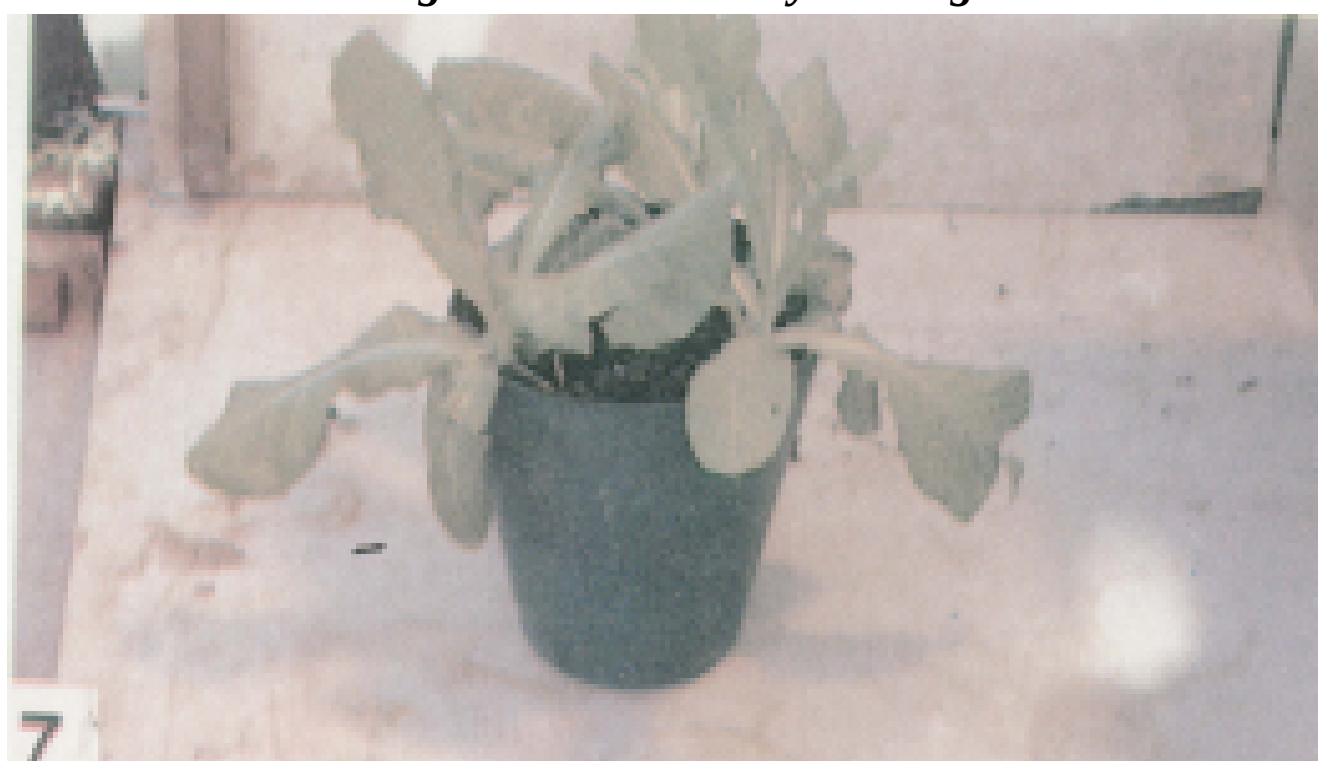


Fig 7 : Lettuce variety Cos

PLATE 4



Fig 8 : Onion bulb variety Pusa Red in peat moss media

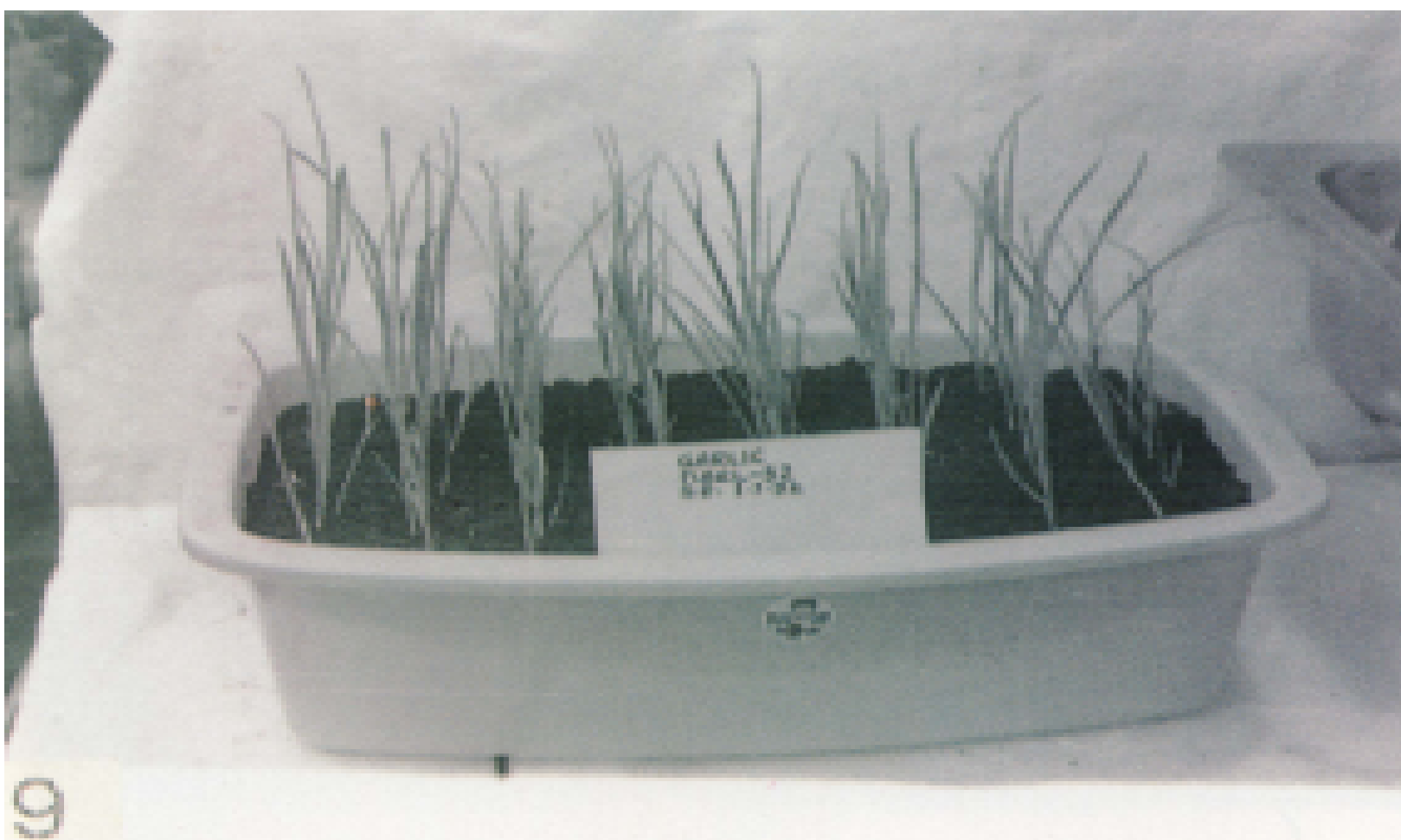


Fig 9: Garlic variety DARL-52

PLATE 5



Fig 10 : Bittergourd variety Pusa Do Mausami.



Fig 11 : Bittergourd variety Pusa Do Mausami in fruiting stage (shown by arrow).

PLATE 6



Fig 12: Bottlegourd variety PSPL in peat moss media



Fig 13: Bottlegourd variety PSPL in Antarctica soil media

whereas in Antarctic soil plant growth and flowering were recorded as very poor (Fig 15).

Coriander and beet root were ready for consumption on 26th day, spinach (palak Banarasi) on 30th day, spinach (Past Hari) on 38th day, fenugreek on 32nd day, lettuce coss on 30th, chinese yellow on 28th, great lake on 24th (Fig 5), cabbage and coss (Figs 7 and 8) on 36th and Parseley (moss caurled) on 40th day of sowing in peat moss media. Data recorded on number and size of fruit or pod and plant height on 51st day of sowing is given in Table VII.

Maximum pod number (22.5) per plant was recorded in Master piece variety (Fig 22) followed by High Altitude selection variety of bean (Table VII) whereas minimum pod number was recorded in VL Baumi variety of bean. Performance of Pusa Sanyog variety was observed best with eight number of fruits and early flowering and fruiting. The fruit size recorded 3.5 cm on 51st day (21st February 1992), increased to 30 cm on the 65th day of sowing. Pod size of bean was recorded maximum 9.0 cm in Pant Anupama variety followed by Master piece (8.4 cm) and Pusa Parvati variety (8.0 cm). Flowering was observed on 51 st day of sowing in PSPL variety of bottlegourd (Fig 12) whereas on 45th and 49th day in Pusa Domausami and 2 ft. long Jaunpuri variety of bittergourd respectively. However, Pusa Domausami variety of bittergourd could only fruit on 48th day of sowing (Figs 10 and 11). Bottlegourd sown in Antarctic soil had poor growth (Fig 13). Flowering was observed on 41st day of sowing in leguminous grass viz., lucern (Fig 19). However, cocks foot grass (*Dactylis glomerata*) did not flower upto 51st day of sowing (Fig 21). Average plant height (in cm), number of leaves per plant, yield per plot and per ha and days taken to harvest fenugreek, beet root, lettuce, spinach and coriander are given in Table VIII. The growth of leafy vegetables viz., spinach, fenugreek, lettuce was very promising unlike the observation made by Audus (1963) and Joshi and Banerjee (1988). The yield of fenugreek (Pusa Kasuri), beet root (Detroit dark red and red ball), lettuce (Chinese yellow and great lake), spinach (Pusa Hari and Palak Banarasi) and coriander (Bulgarian) was recorded as 377.90, 290.69, 256.78, 232.53, 310.07, 368.21, 281.0 and 27.13 q/ha, respectively. Fenugreek was harvested twice for consumption on 32nd and 40th day of sowing whereas beet root (Fig 4) thrice on 26th, 38th and 43rd day. Chinese yellow variety of lettuce (Fig 5) was harvested thrice on 30th, 34th and 40th day, Great lake variety of lettuce thrice on 28th, 32nd and 38th day, Pusa hari variety of spinach (Fig 1) thrice on 30th, 33rd and 37th day, Palak Banarasi variety of spinach thrice on 30th, 34th and 43rd day and Bulgarian variety of coriander (Fig 3) on the 26th day of sowing and after that at every two days interval. Brinjal sown on 31st December 1991 was observed to be poor in growth whereas the PPL variety of brinjal sown during previous austral winter



Fig 14: Fruiting in Australia Green variety of summer squash in peat mass media



Fig 15: Bean vareity Asparagus Black seeded in Antarctic Soil media

Comparative Performance of Thirty Two Crops Grown...
PLATE 8



Fig 16: Bean variety Pusa Cheese Contender and Pant Anupama in fruiting stage grown in peat moss media



Fig 17: Bean variety Pusa Cheese Contender in fruiting Stage grown in Antarctic moss media

PLATE 9



Fig 18: Brinjal variety PPL in fruiting stage grown in peat moss media



Fig 19: Lucern grass in flowering stage grown in peat moss media

PLATE 10



Fig 20: Performance of mint in Antarctica

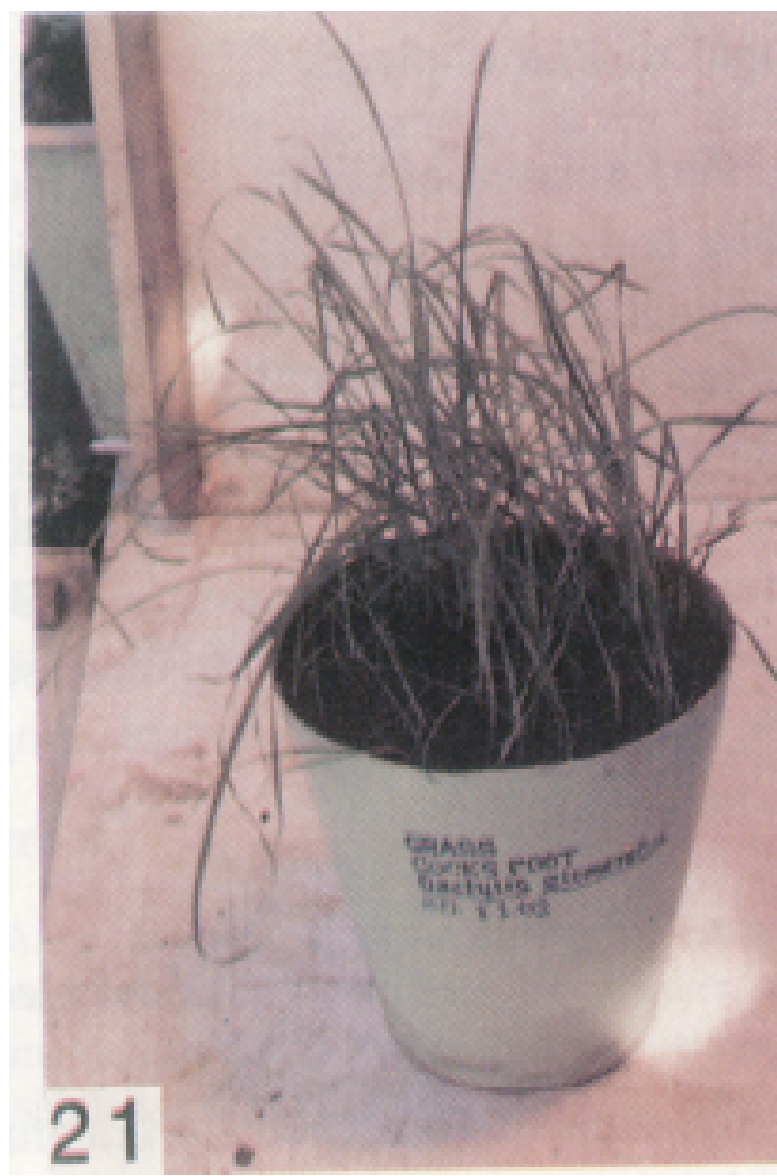


Fig 21: Grass variety Dactylis glomerata in peat moss media



Fig 22: Performance of Master Piece variety of bean in peat moss media

fruited normally (Fig 18). Signs of etiolation and poor vegetable growth reported by Went (1959) was not observed in the present experiments except in the cruciferous plants.

Performance of onion (Fig 8) and garlic (Fig 9) is given in Table VIII. Green leaves of garlic (DARL-52) and onion (Nasik Red) grew ready for consumption on 26th day of sowing. Average length of leaf of garlic and onion was recorded 30.5 and 32.5 cm, respectively and the yield of these vegetables was recorded 145.34 and 339.14q/ha, respectively. Cutting of Himalayan mint (Fig 9) collected from Almora was transplanted on 1 January 1992. Average leaf area was found increasing upto 30th day after which it decreased. The growth rate of the total leaf area increased by 26,56,116 and 157 times on 10th, 20th, 30th and 40th day of transplanting.

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